

Encoder and Decoder - Quick Notes

Encoder

An encoder converts input data into a compact representation (latent vector or embeddings). In machine learning and deep learning, encoders extract important features from the input.

Examples:

- Text Encoder: Converts text into embeddings.
- Image Encoder: Extracts visual features from images.

Decoder

A decoder takes the encoded representation and reconstructs or generates the desired output. It translates latent information back into a readable or usable form.

Examples:

- Language Decoder: Generates text from embeddings.
- Image Decoder: Reconstructs images from encoded features.

Encoder-Decoder Architecture

Widely used in machine translation, summarization, speech recognition, and modern AI models. The encoder processes the input, and the decoder generates the output based on the encoded information.